Review Questions

07/10/2026 Aryan Singh

1. What is an IMU?

Ans. IMU stands for Inertial Measurement Unit. It is a sensor unit that helps measure the movement and rotation of a body such as a rover, drone, robot, or vehicle.

- Accelerometer - measures linear acceleration along the X, Y and Z axes.
- Gyroscope - measures angular velocity, which tells us how quickly the object is rotating around the X, Y and Z axes.

A basic IMU is usually called a 6-DOF IMU because it has 3 accelerometer axes and 3 gyroscope axes. A 9-DOF IMU also has a 3-axis magnetometer, which senses the surrounding magnetic field and helps with heading. For a rover, IMU data can help identify whether the rover is tilting, rotating, or changing its orientation. In practice, IMU data is normally combined with other sensors and estimation methods instead of being treated as perfect position data.

2. What do you mean by yaw, and how does a 9-DOF IMU reduce yaw error?

Ans. Yaw is the rotation of an object about its vertical axis. For a rover, it is basically the direction the rover is facing.

- Roll - tilting left or right.
- Pitch - tilting forward or backward.
- Yaw - turning left or right.

For example, if a rover turns from facing north to facing east, its yaw changes by about 90 degrees. A gyroscope measures angular velocity, not absolute orientation, so yaw is estimated by integrating the gyro readings over time. Small errors and bias can build up, causing gyro drift, so the estimated yaw can slowly become inaccurate even when the rover stops rotating. A 9-DOF IMU adds a 3-axis magnetometer, which senses Earth's magnetic field and gives an approximate heading reference. This reference can be combined with gyro data through sensor-fusion methods to reduce yaw drift.

3. What do you mean by /etc/netplan?

Ans. /etc/netplan is a directory in Linux, not a command. The ‘/’ part represents the root directory, ‘etc’ contains system-wide configuration files, and ‘netplan’ is the directory where Netplan network configuration files are normally stored on supported Ubuntu/Linux systems.

These YAML configuration files can define network interfaces, IP addresses, DHCP, routes or gateways, DNS settings, and Wi-Fi configuration.

4. What does /etc/netplan do in a Linux distro?

Ans. /etc/netplan stores the configuration that tells Netplan how the system network should be set up. The YAML files can be used for Ethernet, Wi-Fi, DHCP, static IP addresses, gateways, DNS-related settings, network bridges, and VLANs.

Netplan provides a high-level way to define network settings in YAML and can work with backends such as systemd-networkd and NetworkManager. After a change, commands such as 'sudo netplan generate', it generates the backend configuration, while 'sudo netplan apply' applies it. Netplan try can be useful for remote changes because it can roll back the temporary configuration if it is not confirmed. Therefore, it's a network configuration tool that translates YAML configuration into configuration understood by a backend.

5. Is a cloud service a virtual machine? Are all cloud services virtual machines?

Ans. No. A virtual machine (VM) is a virtualized computer running on physical hardware, while a cloud service is a service delivered through cloud computing over internet. Cloud services can provide computing, storage, software, databases, networking, or other resources.

Virtual Machine	**Cloud Service**
A virtualized computer	A service provided through cloud infrastructure
Has virtual CPU, RAM, storage and network	Can provide software, platforms, storage, databases or compute
Runs an operating system	May or may not expose a VM to the user
Can exist outside the cloud	Commonly delivered using cloud infrastructure
One type of technology/resource	A broader concept

So, not all cloud services are VMs. For example, IaaS can provide cloud VMs where we manage the OS and applications, while PaaS provides a platform for deploying applications, where we deploy our application, provider manages much of the underlying infrastructure and SaaS provides complete software applications for users, we simply use the software.

6. What are the advantages of Docker?

Ans. Docker is a platform for packaging and running applications in containers. A container keeps the application and the things it needs to run together in an isolated environment.

- Portability - the application and its dependencies are packaged together, so it is easier to move between systems.
- Lightweight - containers share the host OS kernel, so they are generally lighter than full virtual machines.
- Isolation - different applications can run in separate containers with their own dependencies.

- Fast startup - containers can start quickly because they do not need to boot a complete guest OS.
- Dependency management - libraries and configuration can be packaged with the application.
- Easy deployment - the same container image can be used for development, testing, and deployment, making environments more consistent.
- Scalability - additional containers can be launched when more application instances are needed. Larger-scale orchestration is commonly handled by tools such as Kubernetes.

Overall, Docker makes it easier to package, isolate, deploy, and scale applications in a consistent environment.